

Stability Analysis of PLL-Type Grid-Connected Converter Based on The Perturbation Method

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Abstract—In recent years, the widespread adoption of phase-locked loop type grid-connected converters has presented new challenges for transient stability and small-signal stability analysis of power systems. Therefore, it is crucial to develop effective methods for analyzing the stability of converters-interfaced power systems. In this paper, a generalized swing equation model of a single converter connected to an infinite bus is derived. Then, approximate analytical solutions for the model are obtained. The accuracy of the analytical solutions is verified through simulation analysis. Based on the analytical solution, a parameter stability analysis is performed to gain further insights on both transient and small-signal stability. Finally, the proposed results are verified using several examples, demonstrating the accuracy and effectiveness of the suggested methods.

Index Terms—Perturbation method, phase-locked loop, renormalization group, small-signal stability, stability criterion, transient stability, voltage source converter.

I. INTRODUCTION

VOLTAGE source converters (VSCs), connected to the grid through phase-locked loops (PLLs), have become a popular interface for integrating renewable energy sources into modern power systems [1], [2]. However, the stability and transient behavior of modern power systems is becoming increasingly challenging due to the low inertia and fast, complex dynamic processes of VSCs [3]. Several incidents, such as the “8-9” blackout in the U.K. [4] and the sub-synchronous resonance of the Hami wind farm in China [5], highlight the importance of improving the transient stability and small-signal stability of VSC-interfaced power systems.

The instability of a VSC connected to an infinite system can take a variety of complex forms [6]. The focus of this work is on the so-called synchronous stability, which is mainly introduced by the interactions between PLL and the grid. However, the results presented can also be useful when studying other forms of instabilities in power systems. Synchronous stability

incidents related to PLLs have been reported by the industry and academic community. Currently, the main methods for studying synchronous stability are as follows.

- 1) Time domain simulation method [5]. When applying the method, one solves the differential algebraic equations (DAE) of the VSC system through traditional numerical integration, and determines the dynamic stability of the system based on angle trajectories.
- 2) Phase diagram method. The synchronous instability phenomenon of PLLs has been analyzed through the phase diagram of a second-order motion equation of the PLL in [7]. The synchronous stability of PLLs has been studied based on the phase diagram method in [8] and discovered that simply increasing the damping ratio cannot cement the transient stability of VSCs. An adaptive control method of PLL was put forward to enhance the overall transient stability of the VSCs in [9].
- 3) Energy function method. An energy function model of VSC was established in [10], [11]. A fast synchronous stability assessment method for grid-connected VSC was proposed based on the energy function in [12].
- 4) Equal area criterion (EAC). EAC can analyze the transient stability of traditional single-machine infinite systems effectively. Due to the inherent similarity of the synchronization mechanism, it has been extended to the study of transient synchronization stability of converters, which can reveal the mechanism of transient instability occurring in PLL in a more intuitive way. In [13], [14], [15], a large-signal stability analysis based on the EAC was carried out under the weak network connection condition of VSCs. A simple variable-structure method was proposed to improve the resynchronization capability of PLL based on the modified EAC in [16].

Due to the high-dimensionality of inverter-based resources, using time domain simulation to analyze the dynamics of the systems is now considered the predominant choice. Although the system stability can be identified after obtaining the system trajectories by time domain simulation and phase diagram method, it is challenging to provide more insights. The drawback of energy function method and EAC is that they may not yield reliable prediction in VSC stability analysis when damping factor is taken into consideration.

The set of approximate analytical methods is a useful tool for modeling and solving non-linear differential equations [17], [18]. The time-domain expression of the dynamics equation of the PLL had been derived based on the averaging method in [19].

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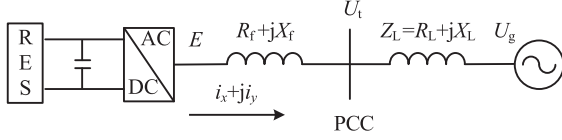


Fig. 1. Diagram of SVIB system.

Another method is the perturbation method. The perturbation method is based on the conception that the non-linearity of the system is insignificant, and thus it is best suited for studying the behavior of non-linear systems not far away from the stable equilibrium point. The perturbation method can obtain the analytical expression of the system dynamics and clarify the influence of system parameters on the system dynamics, which is useful when combined with a time-domain approach. Thus, the perturbation method can allow us to understand the behavior of the system under study.

Inspired by the perturbation method, in this work we propose an analytical method for analyzing the stability of the PLL-type grid-connected converter. Firstly, the generalized swing equation model of the single VSC infinite bus system (SVIB) is derived. Then the approximate analytical solution of the VSC in the transient process is obtained based on the direct perturbation method and the renormalization group method. Based on the expression of the approximate solution, the influence of system parameters on the transient stability and small-signal stability of a VSC is examined. Finally, the validity and correctness of the findings are demonstrated with EMT simulations using MATLAB/Simulink platform.

II. MODELING OF VSC-CONNECTED SYSTEM

This section describes the model setup of SVIB and provide its dynamics model necessary for further study.

A. Description of The VSC-Connected System

Fig. 1 depicts the SVIB system. It shows the internal potential E , voltage U_t at the point of common coupling (PCC), line current $i_x + ji_y$ in the synchronous rotation x - y coordinate system, filter impedance $R_f + jX_f$, line impedance $Z_L \angle \theta_Z = R_L + jX_L$, and the equivalent voltage U_g of the infinite bus.

It is well-known that the response speed between PLL and current controls varies significantly different in most VSCs. The PLL bandwidth typically falls within the range of tens of Hz, enabling rapid tracking and filtering capabilities. However, the bandwidth of current control in VSCs is typically a few hundred Hz [13].

On account of the large time scale separation between the PLL and the current control, the dynamic regulation process of the current control can be simplified by considering the current to be equal to the reference values when only the dynamics of the PLL control time scale are considered.

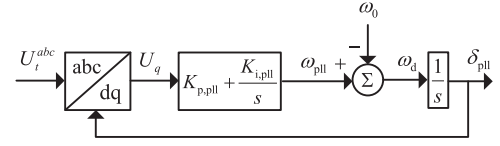


Fig. 2. Control block diagram of PLL [20].

B. The Generalized Swing Equation for a VSC

When the aforementioned assumptions hold, it is possible to eliminate the algebraic variables and obtain a second-order differential equation for a single VSC connected to an infinite bus system. The expression for the q -axis component of the voltage at the PCC point U_q can be derived by transforming the line equation into the coordinate system:

$$(U_d + jU_q)e^{j\delta_{pll}} = U_g + \left(R_L + j\frac{\omega_{pll}}{\omega_0} X_L \right) (I_d + jI_q)e^{j\delta_{pll}} \quad (1)$$

where, U_d and U_q denote the PCC point voltage on the d - q axis, δ_{pll} denotes the difference between the PLL output angle and the infinite bus voltage angle, ω_{pll} denotes the PLL output angular frequency, ω_0 denotes the synchronous angular frequency, and $i_d + ji_q$ denotes the line current in the d - q coordinate system. By separating the real and imaginary parts of (1), one obtains:

$$\begin{cases} U_d = U_g \cos \delta_{pll} + R_L I_d - \frac{\omega_{pll}}{\omega_0} X_L I_q \\ U_q = -U_g \sin \delta_{pll} + R_L I_q + \frac{\omega_{pll}}{\omega_0} X_L I_d \end{cases} \quad (2)$$

The PLL model based on simultaneous rotational coordinate transformation is demonstrated in Fig. 2, and the mathematical model is as follows:

$$\begin{cases} \frac{d\delta_{pll}}{dt} = \omega_{pll} - \omega_0 = \omega_d \\ \frac{d\omega_d}{dt} = K_{p,pll} \frac{dU_q}{dt} + K_{i,pll} U_q \end{cases} \quad (3)$$

where ω_d denotes the difference between the PLL output angular frequency and the synchronous angular frequency, $K_{p,pll}$ and $K_{i,pll}$ denote the proportional and integral coefficients of PI control.

Equation (2) is substituted into PLL equation and (3) is rewritten into the form of the generalized swing equation [21]:

$$M_{eq} \ddot{\delta}_{pll} + D_{eq} \dot{\delta}_{pll} = P_{eq} - U_g \sin \delta_{pll} \quad (4)$$

where M_{eq} , D_{eq} and P_{eq} denote the equivalent time constant of inertia, equivalent damping coefficient, and equivalent mechanical power of PLL, respectively, with the following expressions:

$$\begin{aligned} M_{eq} &= \frac{1}{K_{i,pll}} \left(1 - \frac{K_{p,pll} X_L I_d}{\omega_0} \right) \\ D_{eq} &= \frac{K_{p,pll}}{K_{i,pll}} U_g \cos \delta_{pll} - \frac{X_L I_d}{\omega_0} = \alpha \cos \delta_{pll} - \beta \\ P_{eq} &= R_L I_q + X_L I_d \end{aligned} \quad (5)$$

where

$$\alpha = \frac{K_{p,pll}}{K_{i,pll}} U_g, \beta = \frac{X_L I_d}{\omega_0}. \quad (6)$$

Comparing the generalized swing (4) with the synchronous machine swing equation highlights that the equivalent damping in the former changes with the angle of the grid voltage when operating within normal levels. However, under certain conditions, the equivalent damping of PLL becomes negative.

During a severe voltage dip, when the U_g voltage drops to 5-10% of its normal range, the PLL equation represented by the generalized swing equation changes its output current reference due to the low voltage ride-through control (LVRT), leading to a significant drop in the equivalent damping coefficient. On this occasion, the undamped swing equation can be approximated to simulate the PLL dynamics.

During the LVRT mode, generating reactive power is the priority of a VSC and the output of active power is therefore limited. Therefore, the reference current during the LVRT mode can be expressed as

$$\begin{cases} I_{d,lvr} < \sqrt{I_{\max}^2 - I_{q,lvr}^2} & U_g < 0.9 \\ I_{q,lvr} = K_u(0.9 - U_g) \end{cases} \quad (7)$$

where $I_{d,lvr}$ and $I_{q,lvr}$ denote the reference current during the LVRT mode, $I_{\max}$ denotes the maximum current of a VSC, K_u denotes the gain of reactive current control.

III. ANALYTICAL ANALYSIS BASED ON THE PERTURBATION METHOD

For a system:

$$\begin{cases} \dot{x} = f(x, \varepsilon, t) \\ x(t_0) = \zeta(\varepsilon) \end{cases} \quad (8)$$

where f denotes a smooth function, and $\zeta(\varepsilon)$ denotes the initial state which is dependent smoothly on ε . For a sufficiently small $|\varepsilon|$, the perturbation method aims to construct an approximate solution that are valid.

It should be noted that, as with all approximate analytical methods, the results obtained using the perturbation method are only approximate and may not be entirely accurate. However, it remains a useful tool for gaining insight into the behavior of non-linear systems.

As a system without an equilibrium point is inherently unstable, we limit our analysis to the scenario where the system has an equilibrium point in this study. The equilibrium point of the system described in (4) was as follows:

$$\delta_0 = \arcsin \frac{P_{eq}}{U_g}. \quad (9)$$

Moving the equilibrium point to the origin:

$$x = \delta - \delta_0 \Rightarrow \delta = x + \delta_0, \quad (10)$$

then (4) is equivalent to:

$$M_{eq} \ddot{x} + (\alpha \cos(x + \delta_0) - \beta) \dot{x} = P_{eq} - U_g \sin(x + \delta_0). \quad (11)$$

Expanding the trigonometric function in (11) and rearranging the equation results in:

$$\begin{aligned} \ddot{x} + \left(\frac{\alpha \sqrt{U_g^2 - P_{eq}^2}}{M_{eq} U_g} \cos x - \frac{\alpha P_{eq}}{M_{eq} U_g} \sin x - \frac{\beta}{M_{eq}} \right) \dot{x} \\ = \frac{P_{eq}}{M_{eq}} - \frac{P_{eq}}{M_{eq}} \cos x - \frac{\sqrt{U_g^2 - P_{eq}^2}}{M_{eq}} \sin x. \end{aligned} \quad (12)$$

The following variables are defined:

$$\begin{aligned} a = \frac{\sqrt{U_g^2 - P_{eq}^2}}{M_{eq}}, b = \frac{P_{eq}}{M_{eq}} \\ c = \frac{\alpha}{U_g} = \frac{K_{p,pll}}{K_{i,pll}}, d = \frac{\beta}{M_{eq}} = \frac{X_L I_d}{\omega_0 M_{eq}}. \end{aligned} \quad (13)$$

Then (11) is abbreviated as:

$$\ddot{x} + (ac \cos x - bc \sin x - d) \dot{x} = b - b \cos x - a \sin x \quad (14)$$

According to the formula for the series expansion of trigonometric functions:

$$\begin{cases} \sin x = x - \frac{1}{6}x^3 + \frac{1}{120}x^5 + o(x^5) \\ \cos x = 1 - \frac{1}{2}x^2 + \frac{1}{24}x^4 + o(x^5) \end{cases} \quad (15)$$

Neglecting the higher order terms in (15) and considering $1/6$ as a small parameter ε result in the compact form:

$$\begin{cases} \ddot{x} + p\dot{x} + qx = f(x, \varepsilon) \\ x(t_0) = \eta(\varepsilon) \\ \dot{x}(t_0) = \mu(\varepsilon) \end{cases} \quad (16)$$

where p and q denote the constant coefficient, μ denotes the initial state function, and f denotes a polynomial function which has the following form:

$$f(x, \varepsilon) = \varepsilon \sum_i K_i x^i \quad (17)$$

where K denotes the constant coefficient, i denotes the order number of expansion.

In subsequent text, two different perturbation solutions for (16), depending upon which operating mode the equation describes, are presented.

A. Direct Perturbation Method During Normal Operation

For normal operation, if the solution of (16) is convergent, it can be expressed as the infinite series of the variable ε :

$$x = x_0 + \varepsilon x_1 + \varepsilon^2 x_2 + \dots \quad (18)$$

By disregarding higher-order terms of ε and retaining only the first two term, and substituting the result into (16), one obtains:

$$\ddot{x}_0 + \varepsilon \ddot{x}_1 + p(\dot{x}_0 + \varepsilon \dot{x}_1) + q(x_0 + \varepsilon x_1) = f(x_0, x_1, \varepsilon). \quad (19)$$

As ε in the above equation is arbitrary, the necessary condition for the equation to possess a solution is that all coefficients associated with the same power of ε must be equal to zero,

$$\begin{cases} \varepsilon^0 : \ddot{x}_0 + p\dot{x}_0 + qx_0 = 0 \\ \varepsilon^1 : \ddot{x}_1 + p\dot{x}_1 + qx_1 = f(x_0) \end{cases} \quad (20)$$

The solution of the first equation of (20), which called the generating solution [18], can be expressed as:

$$x_0(t) = e^{\sigma t}(C_{10} \cos \omega t + C_{20} \sin \omega t) = C_{10}x_c + C_{20}x_s, \quad (21)$$

where C_{10} and C_{20} denote the constant coefficient associated with the initial value, and $\sigma \pm j\omega$ denote a pair of conjugate solutions of the following characteristic equation:

$$r^2 + pr + q = 0, \quad (22)$$

and

$$x_c = e^{\sigma t} \cos \omega t, x_s = e^{\sigma t} \sin \omega t. \quad (23)$$

The asymptotic solution x_{norm} based on the method of variation of constant is obtained as [22]:

$$x_{\text{norm}}(t) = C_{10}e^{\sigma t} \cos \omega t + C_{20}e^{\sigma t} \sin \omega t + \varepsilon e^{\sigma t} \left(\cos \omega t \int^t \frac{f(x_0)x_s}{\dot{x}_c x_s - \dot{x}_s x_c} d\xi - \sin \omega t \int^t \frac{f(x_0)x_c}{\dot{x}_c x_s - \dot{x}_s x_c} d\xi \right) \quad (24)$$

where

$$\begin{aligned} \dot{x}_c &= e^{\sigma t}(\sigma \cos \omega t - \omega \sin \omega t) \\ \dot{x}_s &= e^{\sigma t}(\sigma \sin \omega t + \omega \cos \omega t). \end{aligned} \quad (25)$$

Having calculated the terms $x_0, x_1, x_2, \dots$, it is necessary to prove that $\sum_{k=0}^{N-1} x_k \varepsilon^k$ is indeed an $o(\varepsilon^N)$ approximation of $x(t, \varepsilon)$. In what follows, we show that, under certain conditions, the above approximate solutions are asymptotically valid.

Theorem 1 [23] - Let $D \subset R^n$ be a domain that contain the origin and:

- 1) f and its partial derivatives with respect to (x, ε) up to order N are continuous and bounded for $(x, \varepsilon, t) \in [0, \infty) \times D_0 \times [-\varepsilon_0, \varepsilon_0]$, for every compact set $D_0 \subset D$; if $N = 1$, $[\partial f / \partial x](x, \varepsilon, t)$ is Lipschitz in (x, ε) , uniformly in t ;
- 2) η and its derivatives are continuous for $\varepsilon \in [-\varepsilon_0, \varepsilon_0]$;
- 3) the origin is an exponentially stable equilibrium point of the nominal system;
- 4) the origin is globally uniformly asymptotically stable for the nominal system.

Then, for each compact set $\Omega \subset \{W_1(x) \leq \rho c, 0 < \rho < 1\}$, there is a positive constant ε^* such that for all $t_0 \geq 0, \eta \in \Omega$, and $|\varepsilon| < \varepsilon^*$, system (8) have a unique solution, uniformly bounded on $[t_0, \infty)$, and

$$x(t, \varepsilon) - \sum_{k=0}^{N-1} x_k(t) \varepsilon^k = o(\varepsilon^N) \quad (26)$$

where $O(\varepsilon^N)$ hold uniformly in t for all $t \geq t_0$.

It is easily to see that the equilibrium point of SVIB is:

$$(\delta_0, \omega_0) = \left(\arcsin \frac{P_{\text{eq}}}{U_g}, 0 \right). \quad (27)$$

At the equilibrium point, the condition for the system to be exponentially stable is

$$D_{\text{eq}} > 0. \quad (28)$$

If the initial state belongs to a compact subset of the region of attraction of this equilibrium point, the approximation calculated (24) will be valid for all $t \geq 0$.

B. Renormalization Group Method During the LVRT Mode

In some cases, the approximate solution obtained by direct perturbation method is valid only in a very short period of time. This is because of the presence of the secular term, that is, an expression that grows without bound due to the factor t multiplying the sine or cosine. It is well-known that the solution should be periodic for small disturbance and therefore an unbounded term should not be part of the solution. For such cases, the renormalization group method is most useful as it avoids the adverse effect of the secular term [24], [25], [26].

During the LVRT mode, based on the equation of D_{eq}

$$D_{\text{eq}} = \frac{K_{\text{p,pll}}}{K_{\text{i,pll}}} U_g \cos \delta_{\text{pll}} - \frac{X_L I_d}{\omega_0} \quad (29)$$

It can be seen that the damping is related to the voltage drop depth. When the voltage drop depth is not deep, such as 0.3-0.7 p.u, the approximate analytical solution can still be solved by direct perturbation method. As (29) indicates, when the voltage U_g drops to a minimum of 5-10% of the normal operating value, the damping level is correspondingly reduced to 5-10% of its normal operating level (see Fig. 7 in Section V). In this scenario, the damping coefficient can be assumed to be equal to zero, and the generalized swing equation becomes:

$$\ddot{x} = b - b \cos x - a \sin x. \quad (30)$$

It turns out that the direct perturbation solution for the above system is no longer valid. Rather a more sophisticated approach is necessary.

To proceed, similarly, considering $1/6$ as a small parameter ε gives the following equation:

$$\ddot{x} + ax = 3b\varepsilon x^2 + a\varepsilon x^3. \quad (31)$$

Based on the direct perturbation method, the approximate-on solution x_{lvrt} is obtained as:

$$\begin{aligned} x_{\text{lvrt}}(t) &= C_{10} \cos \omega t + C_{20} \sin \omega t + \\ &\frac{\varepsilon}{\omega} \left(\sin \omega t \int^t f(x_0) \cos \omega \xi d\xi - \cos \omega t \int^t f(x_0) \sin \omega \xi d\xi \right) \\ &= C_{10} \cos \omega t + C_{20} \sin \omega t + \varepsilon (R(C_{10}, C_{20}) + Q(C_{10}, C_{20})) \end{aligned} \quad (32)$$

where R denotes the secular term, Q denotes the remaining term. It is obvious that the solution to (31) should be periodic and therefore this unbounded term R should not be part of the solution.

In order to eliminate the secular term, in what follows we introduce the renormalization group method. The first step of the method is to define the secular series as:

$$z_p = 1 + \varepsilon z_{1p}(M, t) + \varepsilon^2 z_{2p}(M, t) + o(\varepsilon^2) \quad (33)$$

where $z_{ip}, i = 1, 2, \dots$, denotes the sequence of secular polynomial factors, M denotes a set of constants. Then let us define the near-identity transformation:

$$\mathcal{M}(t, \varepsilon) = M y_p = M[1 + \varepsilon z_{1p} + \varepsilon^2 z_{2p} + o(\varepsilon^2)] \quad (34)$$

where $\mathcal{M}$ denotes the slowly varying amplitude. Or

$$M = \mathcal{M}(t, \varepsilon) z_p^{-1} \quad (35)$$

The next step is to differentiate (35) with respect t :

$$\frac{d\mathcal{M}}{dt} = \mathcal{M} z_p^{-1} \frac{dz_p}{dt}. \quad (36)$$

By substituting the secular series expansion of z_p in (36), the amplitude equation is rearranged as follows:

$$\frac{1}{\mathcal{M}} \frac{d\mathcal{M}}{dt} = \varepsilon \frac{dz_{1p}}{dt} + \varepsilon^2 \left(\frac{dz_{2p}}{dt} - z_{1p} \frac{dz_{1p}}{dt} \right) + o(\varepsilon^3). \quad (37)$$

By integrating with respect to t , we can obtain:

$$\ln \mathcal{M} = \ln \mathcal{M}(0) + \varepsilon z_{1p} + \varepsilon^2 \left(z_{2p} - \frac{z_{1p}^2}{2} \right) + o(\varepsilon^3). \quad (38)$$

Now, let us return to our problem (32). The secular series of (32) is:

$$\begin{aligned} R(C_{10}, C_{20}) = & \left(\frac{3\sqrt{a}}{8} (C_{10}^2 + C_{20}^2) - \frac{\sqrt{a}}{64} (C_{10}^2 + C_{20}^2)^2 \right) \\ & \times (C_{10} \sin \omega t - C_{20} \cos \omega t)t. \end{aligned} \quad (39)$$

To meet the requirements of the renormalization group method, let

$$C_{10} = 2r \cos \phi, C_{20} = -2r \sin \phi \quad (40)$$

where r and ϕ denote the amplitude and angle, respectively.

According to (38), it is easy to obtain a first integral:

$$\ln \mathcal{M} = \ln \mathcal{M}(0) + \varepsilon z_{1p} + o(\varepsilon^2). \quad (41)$$

Reverting to plane-polar coordinates:

$$\mathcal{M} = r e^{i\phi}. \quad (42)$$

Substituting (42) into (41) gives:

$$\begin{aligned} \ln r e^{i\phi} = \ln r(0) e^{i\phi(0)} - \varepsilon j \frac{3\sqrt{a}}{2} r^2 t + \varepsilon j \frac{\sqrt{a}}{16} r^4 t + o(\varepsilon^2) \Rightarrow \\ \begin{cases} r = r(0) + O(\varepsilon^2) \\ \phi = \phi(0) - \varepsilon \frac{3\sqrt{a}}{2} r(0)^2 t + \varepsilon \frac{\sqrt{a}}{16} r(0)^4 t + o(\varepsilon^2) \end{cases} \end{aligned} \quad (43)$$

The first-order approximate solution x_{lvrt} is obtained by returning (43) to (32):

$$\begin{aligned} x_{lvrt}(t) = C_{10} \cos \omega t + C_{20} \sin \omega t + \varepsilon Q(C_{10}, C_{20}) \\ = r e^{j\omega t + \phi} + \varepsilon Q(r, \phi) + \text{cc} \end{aligned} \quad (44)$$

where cc denotes the conjugate of the previous term.

In summary, (24) and (44) give the approximate analytical solution of SVIB during the normal operation and the LVRT mode, respectively.

IV. PARAMETRIC STABILITY ANALYSIS

Similar to the rotor angle stability problem of synchronous machines, the stability analysis of PLL can also be classified into transient stability and small-signal stability. Both the instability

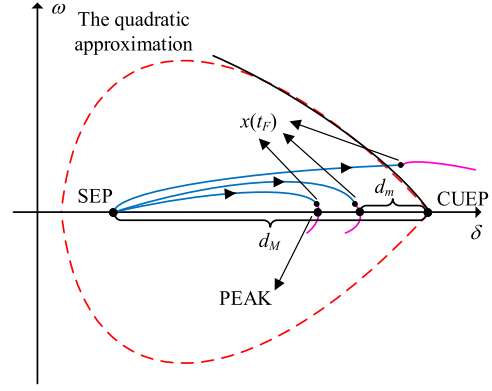


Fig. 3. Stability region for SVIB.

mechanisms and forms of the two kinds of stability are different. Therefore, this section analyzes both of the transient stability and small-signal stability of the VSC.

A. Transient Stability Analysis

One of the key goals of transient stability analysis is to evaluate the ability of the power system to resist large disturbance faults. That is, to judge the stability of the system by calculating the transient stability metric. Whether a power system can maintain synchronization after a large fault can be described as checking whether the state at the clearing time is inside the stability region of the stable equilibrium point (SEP).

The theory for stability region is relatively mature, without getting into too much details, its main conclusion can be summarized as follows.

Theorem 2 [27], [28] – For a smooth autonomous system

$$\dot{x} = f(x), x(0) = x_0. \quad (45)$$

Let ∂A denotes the stability region boundary of the origin, and $W_S(x_i)$ denotes the stable manifold of unstable equilibrium point x_i on the boundary, then under mild conditions

$$\partial A \subset \bigcup_i W_S(x_i). \quad (46)$$

The above celebrated result states that the stability boundary is equal to the set of the stable manifolds of the controlling unstable equilibrium point (CUEP) (refer to Fig. 3 below). Therefore, the transient stability of the system can be determined by the relative distance between the state at the clearing time and the boundary of the post-fault stability region. This is the traditional approach when quantifying the stability margin of synchronous machines.

Theorem 3 [29], [30] – The stable manifold of type-1 UEP has the following quadratic form

$$d_U(x) = [x - x_{uep}]^T \eta + \frac{[x - x_{uep}]^T P [x - x_{uep}]}{2} \quad (47)$$

where η denotes the left eigenvector which satisfies

$$\begin{cases} J^T \eta = \mu \eta \\ \eta^T \eta = 1 \end{cases} \quad (48)$$

where μ denotes the unstable eigenvalue of Jacobin matrix J of the UEP, P denotes the quadratic coefficient which satisfies the following Lyapunov equation

$$CP + PC^T = H \quad (49)$$

where $C = (\mu I_m/2 - J^T)$, I_m denotes the $m \times m$ identity matrix, and

$$H = \sum_{i=1}^m \eta_i H_i \quad (50)$$

where H_i denotes the Hessian matrix of f_i at the UEP.

The stability region boundary theory can be described more intuitively by Fig. 3. In the figure, PEAK denotes the maximum phase angle in the transient trajectory, $x(t_F)$ denotes the system states at the moment when the LVRT mode switches to the normal operation, d_M and d_m denote the distance between SEP and CUEP the distance between PEAK and CUEP, respectively. The red dotted line denotes the stability region boundary of the SEP, while the blue solid line denotes the fault-on trajectory, and the pink solid line denotes the after-fault trajectory, respectively. Besides, the black solid line denotes the quadratic approximation of the stability region boundary of SEP.

Below an alternative way to quantify stability margin of a VSC is proposed. It will be shown subsequently that this stability index is easier to calculate, and more importantly, it appears to have much simpler expression.

Firstly, $d_U(x(t_F))$ is calculated to judge that the system after fault cleared is stable or not. If $d_U(x(t_F)) > 0$ ($d_U(x(t_F)) < 0$) then $x(t_F)$ is inside (outside) the stability region.

Once $d_U(x(t_F)) > 0$, the transient stability metric C_t given in this paper is the phase angle difference between the phase angle at the end of the LVRT mode and the CUEP d_m divides by the phase angle difference between UEP and CUEP d_M , that is:

$$C_t = \frac{d_m}{d_M}. \quad (51)$$

Apparently, the closer the metric is to 1, the more stable the system is. The equivalent damping of the PLL during normal operation is large enough so that ω_d will quickly drop to zero after the fault. Therefore, the peak value of the phase angle of the PLL can be approximately represented by $x(t_F)$.

Assume the fault clearing time is fixed, it can be seen from the approximate expression (44) that the phase angle at the end of the LVRT mode is determined by the constants $r(0)$, $\phi(0)$ and ω .

Firstly, the influence of parameters of the LVRT mode on the $r(0)$ and $\phi(0)$ is analyzed. The solution of the amplitude equation can be calculated as:

$$r(0) \propto (\delta_{\text{norm}} - \delta_{\text{lvrt}}), \phi(0) = 0 \quad (52)$$

where δ_{norm} and δ_{lvrt} denote the system equilibrium points during normal operation and the LVRT mode, respectively. Based on (52), it is concluded that the greater the difference between the system equilibrium points before and during the fault, the greater the amplitude of the PLL phase angle movement during the LVRT mode.

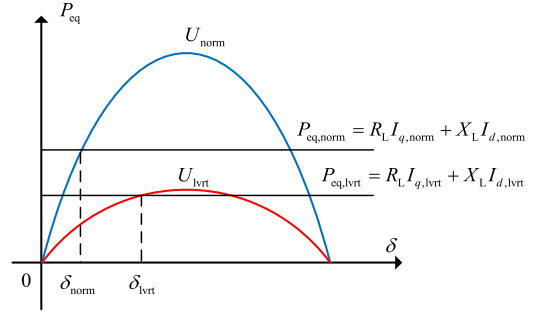


Fig. 4. $P_{\text{eq}}-\delta$ curves.

The equilibrium point is calculated as following:

$$\delta_0 = \arcsin \frac{P_{\text{eq}}}{U_g} = \arcsin \frac{R_L I_q + X_L I_d}{U_g}. \quad (53)$$

Assume that the equilibrium point during LVRT mode exceeds that during normal operation. The active current reference value will increase δ_{lvrt} , resulting in larger PLL phase angle swings during the LVRT mode.

The result can also be explained by Fig. 4, when the distance between δ_{norm} and δ_{lvrt} is farther, the phase angle is more likely to accelerate to move closer to the stable boundary during the LVRT mode, which will lead to instability.

Then, let us study the influence of the parameters on the solution of the characteristic equation during the LVRT mode. During the LVRT mode, the solution of ω is:

$$\omega = \sqrt{a} = \sqrt{\frac{\sqrt{U_g^2 - P_{\text{eq}}^2}}{M_{\text{eq}}}}. \quad (54)$$

It is not difficult to see from (54) that the smaller the equivalent inertia is, the larger the imaginary part of the solution of the characteristic equation is, and the faster the oscillation frequency of the PLL phase angle is during the LVRT mode.

The equivalent inertia is calculated as following:

$$M_{\text{eq}} = \frac{1}{K_{i,\text{pll}}} \left(1 - \frac{K_{p,\text{pll}} X_L I_d}{\omega_0} \right). \quad (55)$$

Apparently, one can see from (55) that the smaller $K_{i,\text{pll}}$ in the PI control parameter is, the larger the equivalent inertia is. So the value of $K_{i,\text{pll}}$ can be appropriately adjusted to slow down the change speed of the PLL phase angle during the LVRT mode.

Therefore, in order to reduce the PLL phase angle swing during the LVRT mode, the operating parameters during the LVRT mode should be adjusted, such as reducing the active power output and reducing the value of $K_{i,\text{pll}}$.

B. Small-Signal Stability Analysis

Damping torque is closely related to the small-signal stability. Insufficient or negative damping torque will cause the real parts of a pair of complex eigenvalues be greater than zeros, resulting in an oscillatory instability.

The solution of the characteristic equation under normal operation is

$$\begin{aligned}\lambda_{1,2} &= \sigma \pm j\omega = \frac{d - ac \pm j\sqrt{(ac - d)^2 - 4a}}{2} \\ &= \frac{\frac{X_L I_d}{\omega_0 M} - \frac{\sqrt{U_g^2 - P_M^2} K_{p,pll}}{M} \pm j\sqrt{(ac - d)^2 - 4a}}{2} \\ &= \frac{\frac{1}{M} \left(\frac{X_L I_d}{\omega_0} - \frac{\sqrt{U_g^2 - P_M^2} K_{p,pll}}{K_{i,pll}} \right) \pm j\sqrt{(ac - d)^2 - 4a}}{2}.\end{aligned}\quad (56)$$

As we expect the system states to return to the stable operating states after a small disturbance, that is to say, the roots of the characteristic equation need to satisfy the condition that the real part is less than 0, that is, the following inequality

$$\frac{X_L I_d}{\omega_0} - \frac{\sqrt{U_g^2 - P_M^2} K_{p,pll}}{K_{i,pll}} < 0. \quad (57)$$

Let C_s denote the criterion of small-signal stability of the system:

$$C_s = \frac{X_L I_d}{\omega_0} - \frac{\sqrt{U_g^2 - P_M^2} K_{p,pll}}{K_{i,pll}}. \quad (58)$$

When $C_s > 0$, the system will inevitably exhibit oscillations and become unstable; conversely, when $C_s < 0$, the system will remain stable after a small disturbance. It is evident from the formula that decreasing $K_{p,pll}$ in PI parameters and increasing $K_{i,pll}$ can lead to instability.

Meanwhile, we can observe the effect of transmission power and transmission distance on the stability of PLL from (58). The increase of transmission power and transmission distance is equivalent to the increase of X_L and I_d , leading to a scenario where the C_s exceeds zero, thereby compromising the stability of PLL.

V. CASE STUDIES

This section provides some case study results to further validate the theoretical findings described in previous sections.

A. Accuracy Verification

1) *Fidelity of the Approximate Analytical Solutions:* In this subsection, the fidelity of the approximate analytical solutions derived above is verified, and the $o(x^3)$ approximation result and the $o(x^5)$ approximation result are compared with the time domain simulation. Here, $o(x^i)$ approximation means the i -th order power series expansion of trigonometric functions.

The simulation is performed using SVIB shown in Fig. 1, with parameters of VSC and the network shown in Table I. where the subscript ‘norm’ and ‘lvrt’ denote the parameter of normal operation and the LVRT mode, respectively. At the moment $t = 0$ s, the infinite bus voltage U_g drops to 0.1 p.u and recovers to 1.0 p.u at $t = 0.15$ s. The results of the time domain simulation,

TABLE I
PARAMETERS OF VSC AND NETWORK

Parameters	$K_{p,pll}$	$K_{i,pll}$	R_L	X_L	U_g
Values	50	4000	0.08	0.4	1.0
Parameters	$I_{d,norm}$	$I_{q,norm}$	$I_{d,lvrt}$	$I_{q,lvrt}$	
Values	1.0	0	0.43	-1.0	

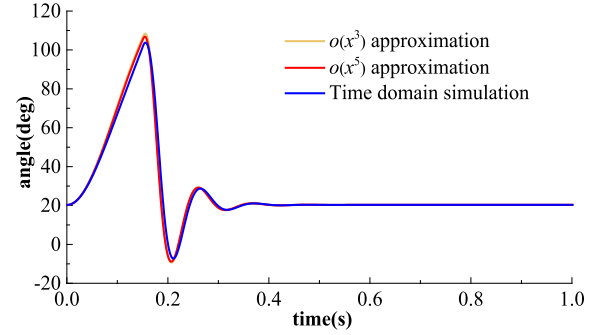


Fig. 5. Results of the time domain simulation, the $o(x^3)$ approximation and $o(x^5)$ approximation.

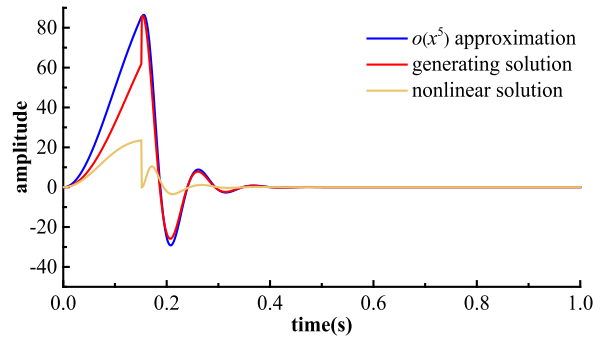


Fig. 6. Structure of the approximate solution.

the $o(x^3)$ approximation and the $o(x^5)$ approximation analytical solution are shown in Fig. 5.

It can be deduced from Fig. 5 that, the $o(x^3)$ and $o(x^5)$ approximation results both have high fitting accuracy in the transient process. But when the PLL phase angle swings to the vicinity of the maximum value, the accuracy of the result of the $o(x^5)$ approximation is obviously higher than that of the $o(x^3)$ approximation. In addition, although the higher-order approximations result may be more accurate, they also consume more computing time. Therefore, it is reasonable to conclude that the $o(x^5)$ approximation can meet the requirements of both accuracy and computational efficiency.

Fig. 6 shows the structure of the $o(x^5)$ approximate solution. The curves in the figure are translated to the origin according to the initial value. It is seen from the figure that the generating solution, obtained from the ε^0 equation in (20), occupies the majority of the approximate solution. The nonlinear solution denotes the solution of the ε^1 equation, which represents the strength of nonlinearity of the system. The amplitude of the curves confirms the hypothesis that the system is weakly nonlinear.

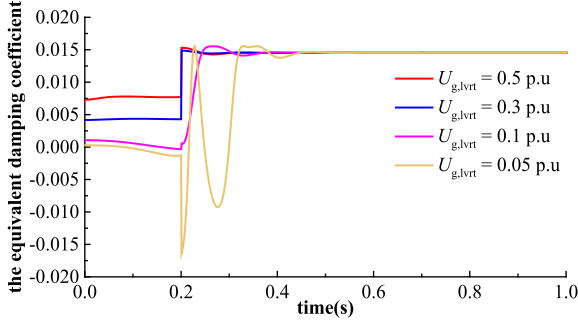


Fig. 7. Variations of damping coefficient at different voltage drop depths.

TABLE II
 C_T CORRESPONDING TO DIFFERENT CASES

Case	time domain simulation		$o(x^2)$ approximation		Errors of $C_T/\%$
	$d_U(x(t_F))$	C_t	$d_U(x(t_F))$	C_t	
1	0.7659	0.4915	0.7879	0.4663	-5.13
2	0.4015	0.3231	0.4657	0.3189	-1.30
3	-0.0943	/	-0.1619	/	/
4	0.3993	0.3157	0.4763	0.3140	-0.51
5	-0.3836	/	-0.0268	/	/
6	0.5322	0.3364	0.6934	0.3766	11.93
7	-0.2548	/	-0.1239	/	/

2) *Condition on Ignoring the Damping Coefficient*: In the subsection, the condition on ignoring the damping coefficient was verified. At $t = 0s$, the voltage U_g drops to different depths (0.5 p.u, 0.3 p.u, 0.1 p.u and 0.05 p.u) and recovers to 1.0 p.u at $t = 0.2s$. The variations of the damping coefficient during the transient process are shown in Fig. 7.

It can be concluded from Fig. 7 that the equivalent damping coefficient during LVRT is much smaller than that during normal operation, especially in severe conditions, i.e., U_g drops to 0.1 p.u and 0.05 p.u. This shows that it is sufficient to ignore the damping coefficient during the LVRT mode.

B. Verification of Stability Criterion

1) *The Influence of Parameters on Criterion*: This subsection examines the influence of different voltage sags, PI control parameter and active current reference on the transient stability metric defined in (51). The parameters of different cases are set as follows (the remaining parameters are consistent with those in Table I).

- 1) Case1: $U_{g,lvrt} = 0.1$ p.u, $I_{d,lvrt} = 0.4$ p.u, $K_{i,pll} = 3000$
- 2) Case2: $U_{g,lvrt} = 0.09$ p.u, $I_{d,lvrt} = 0.4$ p.u, $K_{i,pll} = 3000$
- 3) Case3: $U_{g,lvrt} = 0.08$ p.u, $I_{d,lvrt} = 0.4$ p.u, $K_{i,pll} = 3000$
- 4) Case4: $U_{g,lvrt} = 0.1$ p.u, $I_{d,lvrt} = 0.42$ p.u, $K_{i,pll} = 3000$
- 5) Case5: $U_{g,lvrt} = 0.1$ p.u, $I_{d,lvrt} = 0.44$ p.u, $K_{i,pll} = 3000$
- 6) Case6: $U_{g,lvrt} = 0.1$ p.u, $I_{d,lvrt} = 0.41$ p.u, $K_{i,pll} = 4000$
- 7) Case7: $U_{g,lvrt} = 0.1$ p.u, $I_{d,lvrt} = 0.41$ p.u, $K_{i,pll} = 5000$

At moment $t = 0s$, the infinite bus voltage U_g drops to different values and recovers to 1.0 p.u at $t = 0.23s$. The transient stability metric $d_U(x(t_F))$ and C_t are shown in Table II.

As the voltage sags being deeper in Case1 and Case2, the stability metric gradually decreases, which demonstrates that the system tends to be more unstable. Comparing Case1 and Case4,

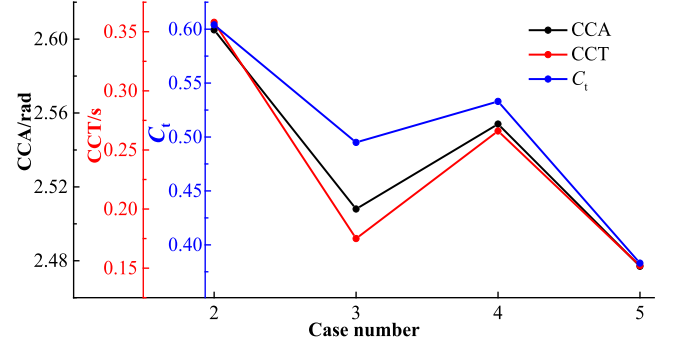


Fig. 8. Comparison with the EAC method.

it follows that as the active reference current increases during the LVRT mode, the stability metric gradually decreases, which means that the stability margin becomes smaller. It verifies that the greater the active current during the LVRT mode, the more likely the system will be transiently unstable. Observed from Case1 and Case6 that the increase of $K_{i,pll}$ also leads to the decreases of stability metric. In Case3, Case5 and Case7, the system becomes unstable, which were indicated correctly by the metric $d_U(x(t_F))$.

2) *Comparison With EAC and Simulation Methods*: In this subsection, a comparison between the proposed method and EAC is presented, the familiar time domain simulation method is used as a benchmark.

EAC is an effective method for calculating the critical clearing angle (CCA). Refer to Fig. 4, let δ_c be CCA, it can be obtained by solving the following equation

$$(U_g - U_{g,lvrt}) \cos \delta_c + (P_{eq,norm} - P_{eq,lvrt}) \delta_c = (\pi - \delta_{norm}) P_{eq,norm} - P_{eq,lvrt} \delta_{norm} - (U_g + U_{g,lvrt}) \cos \delta_{norm} \quad (59)$$

Critical clearing time (CCT) can be obtained by time domain simulation in a trial and error fashion. Both CCA and CCT are reliable metrics for studying the stability of VSC.

We set $K_{p,pll} = 20$ in Case2-Case5, because EAC depends on weak damping. The corresponding CCAs and CCTs are obtained using EAC and time domain simulation methods in Fig. 8, respectively. Meanwhile, the metric C_t proposed in this paper at $t = 0.15s$ is also given in the figure.

As can be seen from the figure, all three methods conclude that the system in Case5 exhibits the poorest performance in terms of transient stability, while the system in Case2 demonstrates the best performance. The comparison presented above shows that in general the results obtained using the afore-mentioned methods are consistent.

The main advantage of our method is that it can be extended to study systems with multiple VSCs in a straightforward way, while the standard EAC does not. In particular, the availability of maximum rotor angle allows one to identify the correct CUEP, a key step in obtaining accurate stability margin estimation. The readers are referred to [31] for some preliminary results.

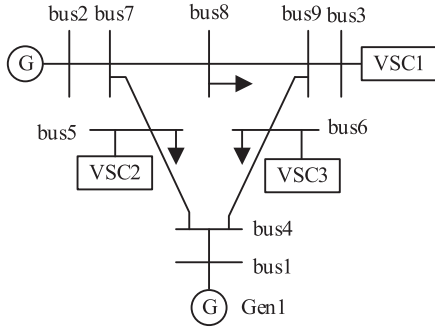


Fig. 9. Modified 3-machine 9-bus system.

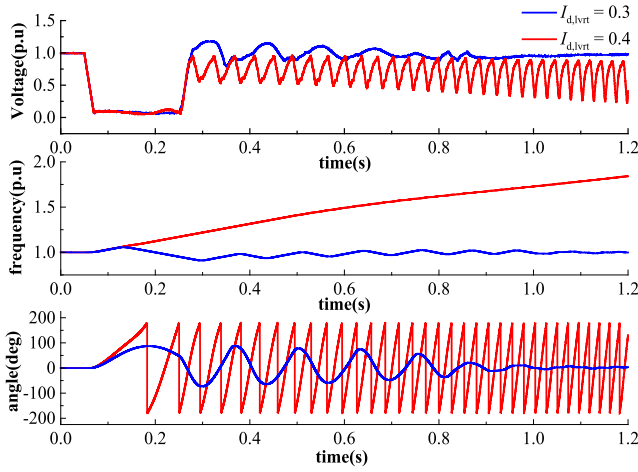


Fig. 10. Voltage of PCC, PLL frequency and phase angle of VSC1 when VSC1 is connected.

C. Verifications Based on Electromagnetic Transient Simulation

The above transient stability results are verified in the electromagnetic transient simulation using MATLAB/Simulink in this subsection. The step size was set to 5×10^{-6} s. The system containing VSCs was obtained by modifying the familiar 3-machine 9-bus system shown in Fig. 9. The synchronous machine on bus3 is replaced with VSCs, and two VSCs are connected to bus5 and bus6 respectively.

Different scenarios are considered, such as different current references, different penetration levels for VSCs. The parameters of VSCs are as shown in Table III.

A short circuit fault is applied between bus8 and bus9, 10 kilometers away from bus9. The fault started at $t = 0.05$ s, and was cleared at $t = 0.25$ s. The penetration level is regulated by controlling the number of interconnected VSCs. The active power references of VSC2 and VSC3 are equal to 0.5 p.u. The simulation results are shown below.

As can be seen from Fig. 10, Fig. 11, and Fig. 12, with the increase of the active current reference during LVRT, VSC1 changes from transient stability to transient instability, which is consistent with the results of the single machine simulation, proving the validity of the results of the paper. It is worth

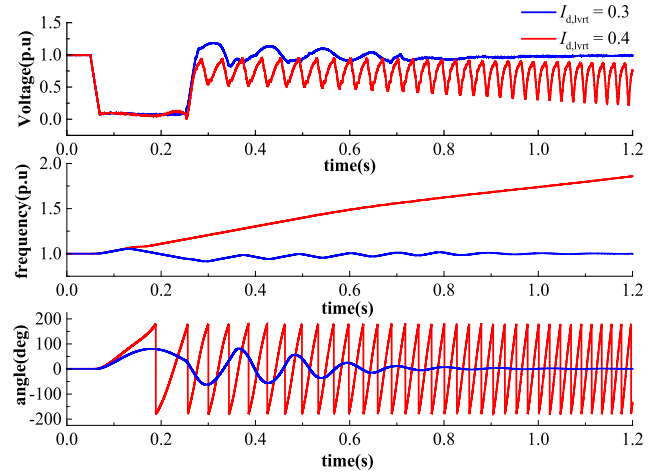


Fig. 11. Voltage of PCC, PLL frequency and phase angle of VSC1 when VSC1,VSC2 are connected.

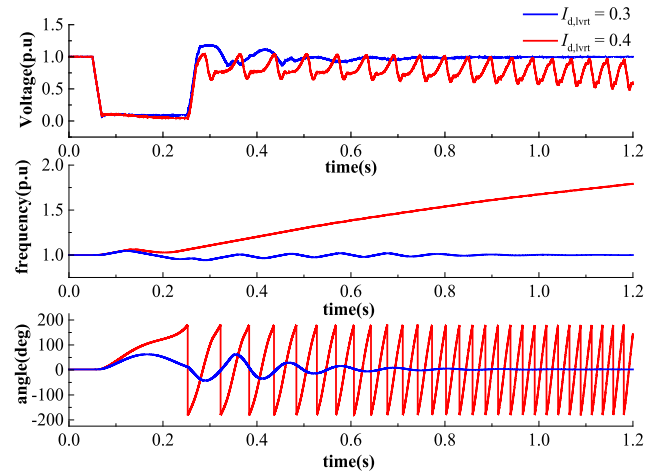


Fig. 12. Voltage of PCC, PLL frequency and phase angle of VSC1 when VSC1,VSC2 and VSC3 are connected.

noting that the same conclusion can be obtained under different penetration levels.

D. Verification of Small-Signal Stability Criterion

1) *Influence of PI Parameter on Stability Criterion:* The validity of the proposed small-signal stability criterion is examined in this subsection. To proceed, we adjust the active current reference value during normal operation to 1.5 p.u. The remaining parameters of the system are consistent with Table I. Then we calculate the small-signal stability metric under different PI parameter combinations, and the results are shown in Table IV.

According to the above calculation results, it can be concluded that the system in Case1 will not experience oscillation instability, and the systems in Case2 and Case3 will experience oscillation instability after a fault occurs. The result can be justified using time domain simulation. The fault is applied at $t = 0$ s, the voltage of infinite bus drops to 0.1 p.u, and recovers to 1.0 p.u at $t = 0.1$ s. The simulation results of the three cases are shown in Fig. 13.

TABLE III
PARAMETERS OF VSC1

Parameters	Values	Descriptions
U_b	575 V	Voltage base
S_b	100 MVA	Power base
f_b	50 Hz	Rated frequency
L_F	0.15 p.u	Filter inductance
C_F	0.05 p.u	Filter capacitor
L_g	0.4 p.u	Inductance of grid impedance
R_g	0.08 p.u	Resistance of grid impedance
U_g	1.0 p.u	Grid voltage amplitude
ω_g	1.0 p.u	power system frequency
P_{ref}/Q_{ref}	1.0 / 0.0 p.u	Active / Reactive Power reference
$K_{p,pll}/K_{i,pll}$	40 / 2500	PI parameters of PLL
$K_{p,c}/K_{i,c}$	1 / 50	PI parameters of current loop
$K_{p,p}/K_{i,p}$	2 / 50	PI parameters of active power loop
$K_{p,q}/K_{i,q}$	0.05 / 5	PI parameters of reactive power loop

TABLE IV
SMALL-SIGNAL STABILITY METRIC OF DIFFERENT PARAMETERS OF PI CONTROL

Case	Parameters of PI control	$C_s / 10^{-4}$	Stable or not
1	$K_{p,pll} = 20, K_{i,pll} = 4000$	-12.8	Yes
2	$K_{p,pll} = 10, K_{i,pll} = 4000$	4.8	No
3	$K_{p,pll} = 20, K_{i,pll} = 7000$	2.3	No

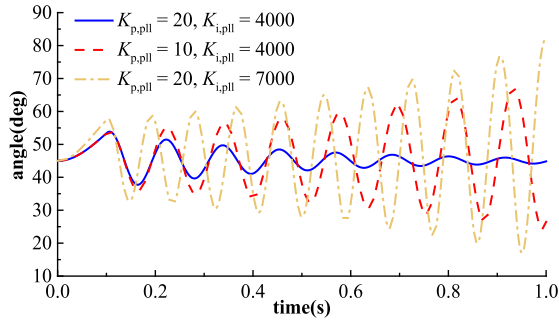


Fig. 13. System dynamics under different PI parameter combinations.

TABLE V
CRITERION FOR DIFFERENT TRANSMISSION DISTANCE AND TRANSMISSION POWER

Case	transmission distance and transmission power	$C_s / 10^{-4}$	Stable or not
1	$I_{d,norm} = 1.8, X_L = 0.4$	-12.8	Yes
2	$I_{d,norm} = 2.2, X_L = 0.4$	4.3	No
3	$I_{d,norm} = 1.5, X_L = 0.55$	0.4	No

As one can see from Fig. 13 that the trajectories of the system under different PI control parameter combinations are consistent with the theoretical findings. This verifies that the small-signal stability criterion proposed in this paper is valid.

2) *Different Transmission Power and Distance:* The influence of transmission power and distance on the stability of the VSC is verified in this subsection. The remaining parameters of the system are consistent with Table I. The small-signal stability criterion under different transmission distances and transmission power combinations is calculated, and the results are shown in Table V.

According to the above calculation results, it can be concluded that the system in Case1 will not experience oscillation instability, and the systems in Case2 and Case3 will experience oscillation instability after a fault occurs. This shows that with the increase of transmission power and transmission distance of a VSC, the PLL is more susceptible to small-signal instability.

VI. CONCLUSION

In this work, a large-signal VSC transient stability analysis method is proposed. A direct perturbation method is described to compute the transient responses of PLL during normal operation, and the renormalization group method is suggested to compute the response during the LVRT mode. It is shown that the integration of the two approaches provides precise analytical solution for the generalized swing equation of PLL. A new phase-domain stability margin, which makes use of the analytical solutions, is defined. The influence of system parameters on the stability of VSC is conveniently analyzed using the suggested stability margin expression. The result of small-signal parametric stability is also provided. The theoretical findings have been further validated with several numerical examples. It is thought that the introduced results can be extended to investigate other forms of VSC instability, this topic remains to be our future work.

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